

## Solution Requirements

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 28 June2026                     |
| <b>Team ID</b>       | XXXXXX                          |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 4 Marks                         |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description                                                                                                                   | Priority (High/Medium/Low) |
|------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1    | Authentication       | <i>Users can securely access the application using login credentials or authenticated sessions (if login is implemented)</i>              | High                       |
| 2    | Authorization levels | <i>Admin can manage the system and view prediction history. Users can submit applications and view only their own prediction results.</i> | High                       |
| 3    | External interfaces  | <i>The system interacts with the trained Machine Learning model, local dataset, Flask backend, and optional SQLite database.</i>          | High                       |

|   |                                   |                                                                                                                                                                          |        |
|---|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|   |                                   |                                                                                                                                                                          |        |
| 4 | Transactions processing           | <i>User inputs are validated, preprocessed, passed to the ML model, and the prediction result is generated instantly.</i>                                                | High   |
| 5 | Reporting                         | <i>Display prediction results, confidence score, model accuracy, and maintain prediction history/dashboard.</i>                                                          | Medium |
| 6 | Business rules, etc.              | <i>Credit card approval is determined based on applicant information such as income, employment, education, credit score, loan history, and other financial factors.</i> | High   |
| 7 | Compliance to laws or regulations | <i>User data must be handled securely, respecting privacy guidelines and protecting sensitive financial information.</i>                                                 | Medium |
| 8 | Other                             | <i>Allow users to make multiple predictions and provide a responsive web interface for easy access.</i>                                                                  | Low    |

**Step 2: Non-Functional Requirements (NFR)**

| S.No | NFR Category | Requirement Description | Target Metric / Acceptance Criteria |
|------|--------------|-------------------------|-------------------------------------|
|      |              |                         |                                     |

|          |                            |                                                                                                                   |                                                            |
|----------|----------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>1</b> | Performance & Speed        | <i>Generate prediction results quickly after user submission.</i>                                                 | <i>Prediction response in less than 2 seconds</i>          |
| <b>2</b> | Scalability                | <i>The application should support increasing users and larger datasets without major performance degradation.</i> | <i>Supports 1,000+ prediction requests efficiently</i>     |
| <b>3</b> | Security & Data Privacy    | <i>Protect applicant information using secure coding practices and encrypted communication where applicable.</i>  | <i>Secure data handling and protected user information</i> |
| <b>4</b> | Reliability & Availability | <i>The system should provide consistent prediction results with minimal downtime.</i>                             | <i>99% application availability</i>                        |
| <b>5</b> | Usability & Accessibility  | <i>Provide a simple, responsive, and user-friendly interface accessible from desktop and mobile devices.</i>      | <i>Easy navigation and responsive design</i>               |
| <b>6</b> | Other                      | <i>Maintainable, modular, and well-documented code to support future updates and deployment.</i>                  | Clean code with proper documentation                       |